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I. INTRODUCTION

$8\pi G_N = 1?$, spacetime signature $(+, -, -, -)$

II. SETUP

A. Lorentzian Regge calculus

Regge calculus [1] is a discrete gravitational theory on simplicial manifolds Δ with the length of edges $\{l_e\}$ dynamical. In a Lorentzian setting, a simplicial manifold consists of intrinsically flat Lorentzian 4-simplices $\sigma \in \Delta$. Curvature is located at triangles $t \in \Delta$ and captured by deficit angles $\delta_t(\{l_e\})$, lying in the space orthogonal to t . For t spacelike (timelike), this space is isomorphic to $\mathbb{R}^{1,1}$ ($\mathbb{R}^2$).¹ Lorentzian deficit angles are defined as $\delta_t^{L,\pm} = \mp i2\pi - \sum_{\sigma \supset t} \psi_{\sigma,t}^{L,\pm}$, following the conventions of [2, 3], where the $\psi_{\sigma,t}$ are dihedral angles. “ $\pm$ ” indicates a choice of sign which becomes important for causally irregular configurations [2–4] discussed below. The bulk action of Lorentzian Regge calculus is given by [1–3]

$$S_R[\{l_e\}] = \sum_{t \in \Delta: \text{tl}} |A_t(\{l_e\})| \delta_t^E(\{l_e\}) + \sum_{t \in \Delta: \text{sl}} |A_t(\{l_e\})| \delta_t^{L,\pm}(\{l_e\}), \quad (1)$$

with $|A_t|$ the absolute value of the triangle area. In the presence of a boundary, $\partial\Delta \neq \emptyset$, S_R attains boundary terms such that overall additivity is ensured. The equations of motion are given by $\frac{\partial S_R}{\partial l_e} = 0$ supplemented by the Schläfli identity, $\sum_t |A_t| \frac{\partial \delta_t}{\partial l_e} = 0$.

Regge calculus can be formulated in terms of different variables [5, 6], with area variables [7–9] playing a particularly important role here due to their connection to LQG and spin-foams. The relation between area and length variables is in general not invertible, and area variables need to be further constrained to yield the dynamical equations of length Regge calculus [3, 6, 9]. However,

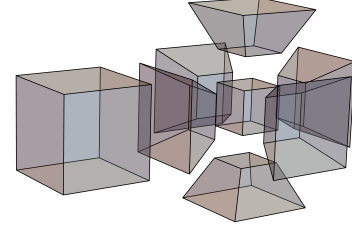


FIG. 1: Boundary of a 4-frustum given by two 3-cubes and six equal 3-frusta.

in the symmetry reduced setting introduced below, the relation of area and length is globally invertible and the aforementioned constraints trivialize. For the remainder, we study the classical dynamics of area Regge calculus which will be used in future work as a comparison to results of the effective spin-foam path integral.

B. Lorentzian 4-frusta

Lorentzian 4-dimensional frusta are ideal to capture spatially flat, homogeneous and isotropic discrete geometries [10, 11]. The boundary of 4-frusta, depicted in Fig. 1, contains two spacelike 3-cubes and six 3-frusta that are either spacelike or timelike. Its entire geometry is captured by the area of squares contained in the 3-cubes, j_0 and j_1 , and the area of the trapezoids contained in the 3-frusta, k . The general notions of Regge calculus introduced above are straightforwardly adapted to this setting. Gluing 4-frusta along boundary 3-frusta in spatial direction and along boundary 3-cubes in temporal direction, one obtains an extended cosmological spacetime of hypercubical combinatorics $\mathcal{X}_V^{(4)}$ with slices labelled by $n \in \{0, \dots, \mathcal{V}_T\}$ discretized by $\mathcal{V}_S^3$ spacelike 3-cubes. We say that a 4-frustum lies in a “slab” between slices n and $n + 1$.

The squared volume of subcells connecting neighboring slices, i.e. 3-frusta, trapezoids and edges, which are detailed in Appendix ??, carries information on the causal structure. If it is positive (negative), the corresponding subcell is timelike (spacelike). In the symmetry reduced setting, the ratio $\frac{|j_n - j_{n+1}|}{4k_n}$ fully characterizes the causal character of subcells suggesting separating the 4-frusta configurations into three sectors summarized in Tab. ??.

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¹ For the remainder, we neglect lightlike edges, triangles and tetrahedra. See [2] for the definition of deficit angles associated to $(d - 1)$ -dimensional null cells.

As discussed in [11–15], two of these sectors I and II exhibit an irregular light cone structure, also referred to as causality violations, located at spacelike sub-cells. Examples of such irregularities are given by the trouser singularity and the yarmulke singularity associated to spatial topology change [16–18]. For causality violations at $(d - 2)$ -dimensional spacelike hinges the imaginary parts of the dihedral angles do not sum up to $\mp i2\pi$, leading to a complex valued Regge action. Choosing “+” in the convention of Lorentzian angles, the amplitude e^{iS_R} is exponentially suppressed for trouser and exponentially enhanced for yarmulke configurations, vice versa for “−” [3]. Without a complexification of the vari-

ables, the Regge equations of motion for causality violating configurations are complex equations with real variables and therefore overconstrained. Moreover, the results of [11, 12] show that in the respective settings, no real solutions exist. Since this applies also to the setting used in this work we henceforth focus on the causally regular sector III.

C. Lorentzian Regge action for spatially flat cosmology

For a single 4-frustum between, say slice n and $n + 1$, the causally regular Regge action is given by

$$S_{\text{III}}^{(n)} = 6(j_n - j_{n+1}) \sinh^{-1} \left(\frac{j_{n+1} - j_n}{\sqrt{16k_n^2 + (j_n - j_{n+1})^2}} \right) + 12k_n \left[\frac{\pi}{2} - \cos^{-1} \left(\frac{(j_n - j_{n+1})^2}{16k_n^2 + (j_n - j_{n+1})^2} \right) \right] - \Lambda V^{(4)}, \quad (2)$$

where Λ is a cosmological constant and $V^{(4)}$ is the 4-volume defined in Eq. (??) as an equation of (j_n, j_{n+1}, k_n) . By construction, the 4-frustum action is additive such that on a discretization $\mathcal{X}_V^{(4)}$ with $\mathcal{V}_S^3$ vertices in spatial direction and $\mathcal{V}_T + 1$ spatial slices, the action is given by $S_R = \mathcal{V}_S^3 \sum_{n=0}^{\mathcal{V}_T-1} S_R^{(n)}$. Additivity of the action per slab implies that the equations of motion for a given geometric quantity, being either j_n or k_n , will only depend on neighboring geometric labels.

In the following, we study the discrete dynamics of spatially flat cosmology as governed by the Regge equations. To that end, we consider first the case of a single time step, $\mathcal{V}_T = 1$, in Sec. III and generalize then to an arbitrary number of time steps in Sec. IV.

III. RECAP: ONE TIME STEP

The classical and quantum dynamics of a single timestep in discrete Lorentzian cosmology have been studied extensively in [4, 12, 13] for spatial curvature with cosmological constant and in [11, 15] for spatially flat spacetimes with a free scalar field. In this chapter we extend these studies to spatially flat cosmology with a cosmological constant, before then focussing on the case of arbitrarily many time steps.

For $\mathcal{V}_T = 1$, the square areas j_0, j_1 are boundary data and only the trapezoid area k_0 is dynamical. Thus, the Regge equations are simply given by

$$\frac{\partial S_{\text{III}}}{\partial k_0} = 0, \quad (3)$$

where due to the Schläfli identity, the first term in Eq. (2) drops out. Notice that this equation is transcendental and can therefore only be tackled numerically.

Vanishing cosmological constant. In the case of $\Lambda = 0$, the Regge action describes a spatially flat vacuum universe. If the boundary data is non-stationary, i.e. $j_0 \neq j_1$, Eq. (3) exhibits no solution for k_0 . In other words, a classical universe can neither expand nor contract if there is no matter present. This is in accordance with the continuum Friedmann equations in vacuum, $(\frac{\dot{a}}{a})^2 = 0$, with a the scale factor, which are solved only by $a = \text{const}$. On the other hand, if $j_0 = j_1$, Eq. (3) is trivially satisfied and the trapezoid area k_0 is undetermined. In fact, the Regge action vanishes, $S_{\text{III}} = 0$ for all k_0 , and the 4-frusta reduce to 4-cuboids discretizing flat space. The arbitrariness of k is a common phenomenon for flat solutions in Regge calculus [19, 20] and signifies the restoration of diffeomorphism symmetry in the discrete [21, 22]. This symmetry amounts to moving spacelike slices (instead of individual vertices), see also the discussion in [23]. Note that for 4-cuboids, $k = \sqrt{j}H$ with H the height of the cuboid which is directly related to the lapse function N in the continuum [11, 12]. The lapse is the gauge variable of the residual diffeomorphism symmetry after symmetry reduction, encoding time reparametrization invariance.

Non-vanishing cosmological constant. For a non-vanishing cosmological constant, $\Lambda \neq 0$, and stationary boundary conditions, $j_0 = j_1$, the only solution to Eq. (3) is $k_0 = 0$, corresponding to degenerate 4-frusta with vanishing height. Thus, a stationary universe for $\Lambda \neq 0$ only exists if no time passes, i.e. if there is no actual time evolution. The more interesting case is $j_0 \neq j_1$ where a non-trivial solution k_0^* for Eq. (3) exists. More precisely, a unique solution k_0^* is found for any $\Lambda > 0$ and any $j_0 \neq j_1$. The on-shell trapezoid area $k_0^*(\Lambda)$ as a function of Λ behaves as a monomial decay $k_0^* \sim c\Lambda^{-\alpha}$ with $c, \alpha > 0$ constants depending on j_0, j_1 . For $\Lambda \neq 0$, the solution diverges and ceases to exist at $\Lambda = 0$ in lines with the discussion above. In the limit $\Lambda \rightarrow \infty$, Eq. (3)

is dominated by the Λ -term, i.e.

$$\Lambda \frac{(j_0 + j_1)^2 k}{4V^{(4)}} \approx 0, \quad (4)$$

solved by $k_0^* = 0$.

Note that the restriction on the sign of Λ is in accordance with the continuum Friedmann equations, $(\frac{\dot{a}}{a})^2 = N^2 \frac{\Lambda}{3}$, where real solutions for the scale factor require $\Lambda > 0$.

For completeness, let us contrast these results with the existence of solutions in the spatially spherical case in Lorentzian [12] and Euclidean [24, 25] signature.² In the former setting, note first that for $\Lambda = 0$, no solutions exist. Furthermore, for $\Lambda > 0$, solutions exist only if $j_0 > j_\Lambda$ and $j_1 > j_\Lambda$, where j_Λ is a Λ -dependent minimal area. This is in accordance with the continuum where the scale factor typically follows a cosh-function, attaining a minimal value a_Λ . In Euclidean signature, solutions exist for $\Lambda = 0$. These correspond, in the continuum, to solutions where $\dot{a} = \text{const}$. For $\Lambda > 0$, solutions exist only if $j_0 < j_\Lambda$ and $j_1 < j_\Lambda$ where j_Λ is the same as in the Lorentzian case, however now representing an upper bound. Again, this behavior agrees with continuum results where a Euclidean universe with positive spatial curvature can only reach a maximum value a_Λ .

IV. ARBITRARY NUMBER OF TIME STEPS

In this section, we study the Regge equations of spatially flat Lorentzian cosmology for an arbitrary number of time steps $\mathcal{V}_T$. This amounts to solving $2\mathcal{V}_T - 1$ coupled transcendental equations for the bulk variables $k_0, j_1, \dots, j_{\mathcal{V}_T-1}, k_{\mathcal{V}_T-1}$ for given boundary data $j_0, j_{\mathcal{V}_T}$ and cosmological constant Λ ,

$$\frac{\partial S_{\text{III}}}{\partial k_n} = 0, \quad \frac{\partial S_{\text{III}}}{\partial j_m} = 0, \quad (5)$$

with $n \in \{0, \dots, \mathcal{V}_T - 1\}$ and $m \in \{1, \dots, \mathcal{V}_T - 1\}$. Since the Regge action S_{III} is additive with respect to the slices, i.e. $S_{\text{III}} = \mathcal{V}_S^3 \sum_n S_{\text{III}}^{(n)}$, Eqs. (5) simplify significantly,

$$\begin{aligned} 0 &= \frac{\partial S_{\text{III}}}{\partial k_n} = \mathcal{V}_S^3 \frac{\partial S_{\text{III}}^{(n)}}{\partial k_n}, \\ 0 &= \frac{\partial S_{\text{III}}}{\partial j_n} = \mathcal{V}_S^3 \left(\frac{\partial S_{\text{III}}^{(n)}}{\partial j_n} + \frac{\partial S_{\text{III}}^{(n+1)}}{\partial j_n} \right). \end{aligned} \quad (6)$$

Thus, it is sufficient to consider the single slab action S_{III} , given in Eq. (3), and its derivatives. An explicit form of these equations is given in Appendix ??.

² The construction of the spatially spherical Regge action differs from the construction here, in particular because there exist only finitely many homogeneous triangulations of the 3-sphere, see [12, 25] for details.

A priori, the number of cubes in each spacelike slice, $\mathcal{V}_S^3$, enters the total Regge action only as a multiplicative factor due to the symmetry restriction. Thus, in the following two sections, we consider $\mathcal{V}_S = 1$ for simplicity. However, let us mention here that in Sec. IV C, the parameter $\mathcal{V}_S$ turns out to be crucial to obtain the correct continuum limit. We tackle the problem of finding solutions to Eq. (5) first analytically in the subsequent section and develop numerical methods thereafter.

Add explicit equations to the appendix, mention simple dependence, additivity

A. Analytical considerations

For vanishing cosmological constant, $\Lambda = 0$, and stationary boundary data, $j_0 = j_{\mathcal{V}_T}$, the results for a single time step obtained in the previous section straightforwardly generalize. That is, if $\Lambda = 0$, the Regge equations are satisfied if and only if the boundary data satisfies $j_0 = j_{\mathcal{V}_T}$. Then, the bulk areas $j_n = j_0$ for all $n \in \{1, \dots, \mathcal{V}_T - 1\}$, and the trapezoid areas are undetermined, marking the restoration of time reparametrization invariance of flat space. On the other hand, for $\Lambda \neq 0$ and boundary data $j_0 = j_{\mathcal{V}_T}$, no non-degenerate solution to the Regge equations is found. A unique solution exist for $j_0 \neq j_{\mathcal{V}_T}$ when $\Lambda > 0$ which is in accordance with the continuum Friedmann equations. To summarize, a spatially flat vacuum universe is flat and stationary while a positive cosmological will necessarily lead to curved non-stationary solutions. For the remainder, we assume $\Lambda > 0$ and $j_0 \neq j_{\mathcal{V}_T}$.

Refer to [11] and explain the monotonicity and continuum limit in the height variable picture.

- monotonicity
- analytical continuum limit

B. Numerical implementation

Solving the Regge equations explicitly requires numerical methods. In this work, we employ the package `NLSolve`³ in `Julia`, designed for solving non-linear equations. Note that while Eqs. (5) define a real-valued function in real variables, `NLSolve` can also be applied to complex numbers which may be relevant for the complex critical point method [1]. The algorithm we developed for this work specifically tailored to Eq. (5) amounts to the `solve_eom`⁴ method which we explain in the following.

³ A documentation of the package can be found at github.com/JuliaNLSolvers/NLSolve.jl. Note that also the `NSolve` method in `Mathematica` can be used, but is expected have lower performance in terms of accuracy and computation time.

⁴ See this github repository for the source code.

First, note that the method `nsolve` of the `NSolve` package takes as input a vector valued function $\vec{F}$, a vector of initial variables $\vec{x}_0$, and parameters for the number of iterations as well as the error tolerance for solutions. If the problem is sufficiently well-behaved, `nsolve` finds a solution to the equation $\vec{F}(\vec{x}) = \vec{0}$ using $\vec{x}_0$ as a starting guess of the solving algorithm. In our case, $\vec{F}: \mathbb{R}^{2\mathcal{V}_T-1} \rightarrow \mathbb{R}^{2\mathcal{V}_T-1}$, with

$$\vec{F} = \left(\frac{\partial S_{\text{III}}}{\partial k_0}, \frac{\partial S_{\text{III}}}{\partial j_1}, \dots, \frac{\partial S_{\text{III}}}{\partial j_{\mathcal{V}_T-1}}, \frac{\partial S_{\text{III}}}{\partial k_{\mathcal{V}_T-1}} \right), \quad (7)$$

$$\vec{x} = (k_0, j_1, \dots, j_{\mathcal{V}_T-1}, k_{\mathcal{V}_T-1}). \quad (8)$$

Therefore, adapting Eq. (5) to be solved by `nsolve` is done in two steps: 1) set up the function $\vec{F}$ for fixed boundary areas $j_0, j_{\mathcal{V}_T}$ and cosmological constant Λ , and 2) propose a starting guess $\vec{x}_0$. While step 1) is straightforward, step 2) requires particular care because the starting guess $\vec{x}_0$ is crucial for convergence and good performance of the solving algorithm. In particular, as $\mathcal{V}_T$ grows, the number of variables and equation grows and the algorithm because more sensitive to a well-chosen $\vec{x}_0$.

By trial and error, one can find sufficient starting guesses for `nsolve` to converge given various small $\mathcal{V}_T$.⁵ For the solutions $\vec{x}^* = (k_0^*, j_1^*, \dots, j_{\mathcal{V}_T-1}^*, k_{\mathcal{V}_T-1}^*)$, the following patterns are being observed. First, the j_n^* can be approximated as

$$j_n^* \approx j_0 + n\Delta j, \quad n \in \{1, \dots, \mathcal{V}_T - 1\}, \quad (9)$$

with $\Delta j = (j_{\mathcal{V}_T} - j_0)/\mathcal{V}_T$. Second, let k_0^* be the solution of the trapezoid area for $\mathcal{V}_T = 1$. Then, for $\mathcal{V}_T > 1$, the k_n^* are approximated by

$$k_n^* \approx k_0^*/\mathcal{V}_T, \quad n \in \{0, \dots, \mathcal{V}_T - 1\}. \quad (10)$$

That is, to find a good initial guess for the trapezoid areas, one only needs to find a solution k_0^* for a single time step. Crucially, for a single equation, `nsolve` converges for a broad range of initial guesses, so k_0^* can be easily found. In summary, finding a good initial guess $\vec{x}_0$ amounts to finding the one time step solution k_0^* and then setting

$$\vec{x}_0 = \left(\frac{k_0^*}{\mathcal{V}_T}, \dots \right) \quad (11)$$

- convergence monitor
- scaling of computation time as function of $\mathcal{V}$

C. Continuum limit

blabla

⁵ Tried using continuum, but didn't work as well.

V. CONCLUSION

Extension to $k = \pm 1$.

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Appendix A: Frusta geometry

Appendix B: Series convergence acceleration

Before we explicitly address the numerical evaluation of the partition function, let us digress briefly to introduce the Shanks transform and Wynn's ϵ -algorithm which have been applied to effective spin-foams⁶ in [13]. In the subsequent sections, we will frequently encounter unbounded sums of the form

$$\mathfrak{S} = \lim_{n \rightarrow \infty} \mathfrak{S}_n = \lim_{n \rightarrow \infty} \sum_{j=1}^n a_j \quad (B1)$$

for some complex sequence $\{a_j\}$. Assuming the partial sum $\mathfrak{S}_n$ to be known and of the form [26]

$$\mathfrak{S}_n = \mathfrak{S} + \sum_{j=0}^{k-1} c_j \lambda_j^n, \quad (B2)$$

with c_j coefficients and the $1 > |\lambda_0| > \dots > |\lambda_{k-1}|$ referred to as transients, there are $2k + 1$ unknowns given by the limiting value $\mathfrak{S}$, the c_j and the λ_j . In order to solve for $\mathfrak{S}$, one utilizes $2k + 1$ consecutive sequence values $\mathfrak{S}_n, \mathfrak{S}_{n+1}, \dots, \mathfrak{S}_{n+2k}$ to obtain the k -th Shanks transform [28, 29] $e_k(\mathfrak{S}_n)$ as a ratio of determinants. For sufficiently large n , the Shanks transform $e_k(\mathfrak{S}_n)$ approximates $\mathfrak{S}$ faster than taking the limit $n \rightarrow \infty$ of the partial sums $\mathfrak{S}_n$.

In [30], Wynn introduced a non-linear recursive relation to efficiently compute the Shanks transform, commonly referred to as ϵ -algorithm. Given the partial sums $\mathfrak{S}_1, \dots, \mathfrak{S}_n$, define [26]

$$\epsilon_{-1}^{(n)} = 0, \quad \epsilon_0^{(n)} = \mathfrak{S}_n. \quad (B3)$$

⁶ A closely related series convergence acceleration, known as Aitken's Δ^2 -process [26] has been applied to infinite bulk variable summations in [27].

Then, ϵ 's of higher k are obtained via the relation

$$\epsilon_{k+1}^{(n)} = \epsilon_{k-1}^{(n+1)} + \frac{1}{\epsilon_k^{(n+1)} - \epsilon_k^{(n)}}. \quad (\text{B4})$$

Wynn has shown [30] that $\epsilon_{2k}^{(n)} = e_k(\mathfrak{S}_n)$ thus providing

a fast algorithm for computing the k -th Shanks transform of a sequence $\mathfrak{S}_n$.⁷

Note that Wynn's ϵ -algorithm can be applied to sequences of more than one variable, which we discuss explicitly in Sec. ??.

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⁷ A Mathematica algorithm for convergence acceleration via Wynn's method can be found in the repository Sequence-

Limit. An implementation in JULIA can be found in <https://github.com/Jercheal/3d-cosmology>.